Kestrel Robotics KR-7 Collaborative Arm - User Manual

Overview and installation

The KR-7 is a six-axis collaborative robot arm designed for light assembly, machine tending and laboratory work. It is intended to work alongside people without a safety cage, provided the risk assessment for the specific task confirms that contact forces stay within acceptable limits. This manual covers installation, basic programming through the teach pendant, maintenance and troubleshooting. It does not replace the separate safety manual, which must be read first.

The arm weighs twenty-two kilograms and must be mounted on a rigid base that will not flex or vibrate during fast moves. A steel table at least twenty millimetres thick, bolted to the floor, is suitable for most uses. Mounting the arm on a mobile cart is possible, but the cart must be braked and the arm's payload settings reduced, because sudden stops can tip a cart that is not firmly anchored.

The control box connects to the arm through a single cable that carries both power and data. Route the cable so that it cannot be pinched by the arm's own motion or stepped on by operators. The control box needs a standard single-phase supply and an earth connection; do not use extension boards or adaptors, which have caused intermittent faults that are hard to trace.

After mounting, run the calibration routine from the pendant's setup menu. The routine moves each joint slowly through its range and records the zero positions. Keep people at least one metre from the arm during calibration, since the arm does not run its collision detection while calibrating and will not stop on contact.

The maximum payload of the KR-7 is 7 kilograms at full reach of 850 millimetres, including the weight of the gripper or any other tool mounted on the wrist flange.

Programming with the teach pendant

Most tasks can be programmed without writing code by guiding the arm by hand and recording waypoints. Press and hold the freedrive button on the back of the pendant, move the arm to the desired position, and press the record key. The arm stores each waypoint with its joint angles and tool orientation, and the sequence can then be replayed, edited or combined with waits and input checks.

Waypoints can be joined by joint moves, which are fastest, or by linear moves, which keep the tool on a straight line. Use linear moves when the tool must enter a fixture or follow an edge, and joint moves for travel through open space. Blending between waypoints smooths the path and shortens cycle time, but a large blend radius can make the arm cut corners close to obstacles, so check blended paths at reduced speed first.

The pendant also supports simple logic: wait for a digital input, set an output, repeat a block a fixed number of times, or branch on a condition. More complex programs can be written in Python on a separate computer and uploaded through the network port, but programs uploaded this way bypass the pendant's step-by-step checks and must be tested carefully in simulation mode before running on the real arm.

Speed in collaborative mode is limited automatically according to the safety configuration. Operators sometimes raise the speed limit to shorten cycle times; any such change must be followed by a new risk assessment, and the change is recorded in the arm's event log together with the user account that made it.

Each joint is limited to a continuous torque of 38 newton-metres, and if a programmed move would exceed this the pendant shows error code J-214 and refuses to run the program.

Maintenance and troubleshooting

The KR-7 needs little routine maintenance. Wipe the arm with a dry cloth weekly and check the cable for wear at the points where it bends. Do not use solvents on the joint covers, which can crack the seals and allow dust inside. In dusty environments such as woodworking shops, fit the optional fabric covers sold for the arm and clean them monthly.

Every six months, run the joint health test from the maintenance menu. The test moves each joint against a known load and compares the motor current with the values recorded when the arm was new. A rising current suggests wear in the gearbox, and the pendant will recommend service once any joint exceeds the warning threshold for two consecutive tests.

If the arm stops unexpectedly during a program, first read the message on the pendant. Protective stops caused by contact clear once the operator confirms the area is safe. Repeated protective stops at the same point in a program usually mean the path passes too close to a fixture, or that the payload setting is lower than the tool actually fitted on the wrist.

Network problems are the most common cause of failed uploads from an external computer. Check that the computer and control box are on the same subnet, and that no firewall blocks the upload port. The control box keeps the last ten uploaded programs, so a failed upload never overwrites a working program.

Gearbox grease in every joint must be replaced after 20,000 operating hours or five years, whichever comes first, and only by a technician certified by Kestrel Robotics.